Agent B Classical Path Report

Work on `op-exposed-hull`, 2026-06-07

Agent B exploration lane

Checkpoint after commit `fa2c032`

Abstract

This document reports the Agent B exploration work on the classical `op-exposed-hull` route performed on 2026-06-07. It is a sandbox report, not a canonical project paper section and not an upgrade of any registry status. The report records the target, all definitions introduced or used by the session, the conjectural proof architecture, the results that were proved at exploration level, the false shortcuts found, the numerical and literature evidence, and the current assessment of promise.

The main conclusion is that the classical route remains promising but is not closed. The work reduces the route to a sharply localized closed-bad-class augmentation problem. Most local finite-dimensional pieces are proof-ready. The current blocker cut is

$$\begin{aligned} C9 &\quad \text{(q/shadow interface)},\\ C12 &\quad \text{(alpha budget or calibrated dual)},\\ C13 &\quad \text{(separated-circuit negative-mass lower bound)}. \end{aligned}$$

The latest C9 campaign killed two tempting shortcuts: distributional q-quasi-closure does not imply rowwise closure, and Step 3 height control does not give $O(\tau)$ leakage for arbitrary shadow witnesses. The surviving C9 target is a coupled averaging law $\pi = m$ controlling both q-flow and the shadow kernel, or else a route to the C12 calibrated-dual obstruction.

Contents

1 Status Vocabulary And Scope

The following status labels are used throughout this report.

canonical A statement already in the canonical registry/report. This report does not change such status.

af validated A canonical statement with an adversarial proof workspace validated by the existing project machinery. No new such result was produced in this session.

ready An Agent B exploration statement with a complete local proof outline, small enough to be proposed to Agent A for canonical packaging. It still needs formulation review, byte-grounding of any external facts, and the Recipe A/B/af pipeline before becoming canonical.

conditional A statement proved modulo a named extra hypothesis.

open A plausible target with no complete proof.

false A tempting statement refuted by an explicit argument or finite counterexample.

evidence Numerical or literature evidence only.

All work reported here stayed in the Agent B lane: `agent-B/**`, `docs/plans/**`, and `docs/worklog.md`. Canonical layers `definitions/`, `argument/`, `proofs/`, and `report/` were not used to promote any new theorem. The underlying canonical open problem remains `op-exposed-hull`.

2 Canonical Starting Point

The project already has a classical reduction in the registry. The relevant open target is the global exposed-hull lemma for exact signed affine retractions. The canonical definition index contains the definitions `def-stochastic`, `def-exposed`, and `def-near-positive-projection`; this report uses them by reference rather than redefining the canonical shards.

Conjecture 1 (Global exposed-hull target, registry `op-exposed-hull`)**.** There are universal constants $c, C, C' > 0$ such that for every exact signed affine retraction P, with rows p_i, satisfying

$$P\mathbf{1} = \mathbf{1}, \qquad P^2 = P, \qquad \operatorname{neg}(p_i) \le \delta,$$

and with $\tau = \sqrt{\delta}$, $\rho = C\tau$, $\kappa = c\tau$, every row satisfies

$$\operatorname{dist}_1(p_i, \operatorname{conv} W_{\rho,\kappa}) \le C'\tau.$$

Here $W_{\rho,\kappa}$ is the set of row vertices of $K = \operatorname{conv}\{p_i\}$ whose exposedness modulus at scale ρ is at least κ.

If Conjecture 1 is proved, the existing cluster-representative classical theorem gives a stochastic idempotent within the sharp $O(\sqrt{\delta})$ scale. Hume's family shows that the square-root scale is the right scale; the session found no evidence that Hume obstructs the exposed-hull formulation.

3 Definitions Used Or Introduced In The Session

The definitions in this section are session-level working definitions. They are included so that later proof packages have a precise vocabulary. They are not canonical definition shards.

Definition 1 (Signed affine retraction)**.** A matrix P is a signed affine retraction if its rows $p_i \in \mathbb{R}^n$ have row sum 1 and

$$P\mathbf{1} = \mathbf{1}, \qquad P^2 = P.$$

Equivalently, each row is fixed by right multiplication:

$$p_i P = p_i = \sum_j p_i(j) p_j.$$

Definition 2 (Negative mass)**.** For a signed probability vector x, set

$$\operatorname{neg}(x) = \sum_{j : x_j < 0} (-x_j).$$

Then $\|x\|_1 = 1 + 2\operatorname{neg}(x)$. The standing small-defect hypothesis in the classical path is $\operatorname{neg}(p_i) \le \delta$ for all rows.

Definition 3 (Sharp parameter)**.** Throughout the session

$$\tau = \sqrt{\delta}.$$

The exposedness and separation scales are

$$\rho = R\tau, \qquad \kappa = k\tau$$

with universal constants chosen in a hierarchy.

Definition 4 (Exposedness modulus)**.** For a row vertex $v = p_a$ and scale ρ, let

$$S_v(\rho) = \{i : \|p_i - v\|_1 \ge \rho\}.$$

The exposedness modulus is
$$e_v(\rho) = \sup_h \min_{i \in S_v(\rho)} h(p_i),$$
where h ranges over affine functions on aff K with
$$h(v) = 0, \qquad 0 \le h(p_i) \le 1 \quad \text{for all rows } i.$$
The well-exposed row-vertex set is
$$W_{\rho,\kappa} = \{v : \ v \text{ is a row vertex and } e_v(\rho) \ge \kappa\}.$$

Definition 5 (Skeleton and bad rows)**.** A skeleton R_0 is a maximal 4ρ-separated subset of $W_{\rho,\kappa}$, and $C = \operatorname{conv} R_0$. A bad row, relative to a threshold $D\tau$, is a row satisfying
$$\operatorname{dist}_1(p_i, C) > D\tau.$$
In the closed-bad-class block the bad set is enlarged by ρ-clusters when vertex language is required.

Definition 6 (Positive-coordinate repair)**.** For each row let $a_i = \operatorname{neg}(p_i)$, write $p_i = p_i^+ - p_i^-$, and define
$$q_i = \frac{p_i^+}{1 + a_i}.$$
Then q_i is a probability vector. The matrix $q = (q_i)$ is the repaired positive-coordinate Markov kernel. For a bad set B, write
$$T = q|_{B,B}, \qquad s_i = q_i(B^c) = 1 - \sum_{j \in B} T_{ij}.$$

Definition 7 (Bad lifetime and fundamental matrix)**.** For a substochastic block $T = q|_{B,B}$, the killed-chain fundamental matrix is
$$N = (I - T)^{-1} = \sum_{m \ge 0} T^m$$
when $\rho(T) < 1$. The quantity
$$\|N\|_{\infty \to \infty}$$
is the maximal expected number of visits to B before exit.

Definition 8 (Separator height and high slices)**.** Let ϕ be an affine separator with l_∞-dual norm at most 1. Set
$$M = \max_j \phi(p_j), \qquad h_i = M - \phi(p_i).$$
The high slice of width γ is
$$H_\gamma = \{i : h_i \le \gamma\}.$$
The two slices used in C9 are
$$H_0 = B \cap \{h_i \le A_0\delta\}, \qquad H_1 = B \cap \{h_i \le G\tau\}.$$

Definition 9 (Shadow witness and shadow kernel)**.** If a vertex u fails $e_u(\rho) \ge \kappa$, a shadow witness is a probability measure σ_u supported on rows at distance at least ρ from u and with controlled separator-height loss. A shadow kernel S on a high bad set H is a substochastic matrix whose row S_i is such a witness, restricted to H. Its row loss is
$$1 - S\mathbf{1}.$$

Definition 10 (Failed-exposedness circuit data)**.** For a non-exposed vertex v, Step 5 produces data

$$\mu \in \Delta(S_v(\rho)), \qquad \alpha_i, \beta_i \geq 0,$$

with

$$\sum_{j \in S_v(\rho)} \mu_j (p_j - v) = \sum_i (\beta_i - \alpha_i)(p_i - v), \qquad \sum_i \beta_i < \kappa.$$

The β-side is controlled. The α-side lies on the zero face of an optimal exposing functional and is the major aggregation obstruction.

Definition 11 (Aggregation parameters)**.** The blocker interaction uses

$$\begin{aligned}
\xi &= \text{q-closure/stationarity error},\\
M &= \text{averaged alpha mass},\\
\theta &= \text{separated witness mass},\\
E &= \text{aggregation residual}.
\end{aligned}$$

The intended closing scale is

$$\xi = O(\tau), \qquad M \leq C/\tau, \qquad \theta \geq c\tau, \qquad E = O(\delta).$$

Definition 12 (Pi-coupled q/shadow closure)**.** The corrected C9 interface asks for a probability law π on a high bad set H such that

$$\pi q(H^c) \leq \xi_q, \qquad \|\pi S - \pi\|_1 \leq \xi_S, \qquad \pi(1 - S\mathbf{1}) \leq \xi_S.$$

The most promising case is $\pi = m$, the occupation or quasi-stationary law produced by the bad-kernel lifetime alternative.

4 Proof Architecture Developed In The Session

The best current architecture is not pure convex geometry. It is

$$\begin{gathered}
\text{maximal exposed skeleton}\\
\longrightarrow \text{ positive-coordinate Markov kernel from } P^2 = P\\
\longrightarrow \text{ bad-kernel resolvent alternative}\\
\longrightarrow \text{ long-lived closed-bad-class alternative}\\
\longrightarrow \text{ LP/circuit augmentation contradiction}\\
\longrightarrow \texttt{ op-exposed-hull}.
\end{gathered}$$

The crucial reason to use the repaired kernel is that every q-step costs only order δ. Therefore an expected bad lifetime of order $1/\tau$ accumulates only order $\delta/\tau = O(\tau)$, which is exactly the target scale.

4.1 Primary Route: Maximal Skeleton And Bad Kernel

Choose R_0 maximal 4ρ-separated in $W_{\rho,\kappa}$. If every row is $O(\tau)$-close to $C = \operatorname{conv} R_0$, the classical cluster theorem finishes. Otherwise form a bad set B. The proof splits:

1. If the bad-to-bad q-block T has

$$\|(I-T)^{-1}\|_{\infty\to\infty} \le C/\tau,$$

   then all bad rows reconstruct from C at $O(\tau)$ cost.

2. If not, finite Markov algebra produces a long-lived q-quasi-closed bad distribution. The remaining task is to prove that such a high bad class contains a new well-exposed vertex far from R_0, contradicting maximality.

4.2 Secondary Route: Robust Coordinates

The robust-coordinate route constructs affine coordinates on selected representatives with row reconstruction $O(\tau)$ and coordinate negative mass $O(\delta)$. Pushing canonical row coordinates through a stochastic kernel preserves $O(\delta)$ negative mass, but the representative interpolation matrix G must satisfy

$$\max_b \sum_a |G_{ba} - \delta_{ba}| = O(\delta).$$

A merely $O(\tau)$ interpolation error exactifies through G^{-1} and can create $O(\tau)$ coordinate negativity, losing the target rate. Numerical LP probes nevertheless found $O(\delta)$ interpolation defect in the tested Hume, quadrilateral, and small-defect exact similarity examples. This route is promising but secondary.

5 Results Proved Or Isolated

No new canonical theorem was proved. The results below are exploration-level proof packages or refutations. Each must still go through Agent A review and the local provenance rules before becoming a registry result.

5.1 Ready Local Lemmas

Label	Status	Statement and proof content
C0 positive-coordinate repair	ready	For $q_i = p_i^+/(1+\mathrm{neg}(p_i))$, $$\left\| p_i - \sum_j q_i(j) p_j \right\|_1 \le 4\delta.$$ This follows directly from $p_i P = p_i$, the sign split $p_i = (1+a_i)q_i - a_i r_i$, and $\|p_j\|_1 \le 1 + 2\delta$.
C1 bad-kernel resolvent	ready	If $G = B^c$ rows are Γ-close to a convex set C, and $L = \|(I-T)^{-1}\|_{\infty\to\infty}$, then every bad row satisfies $$\mathrm{dist}_1(p_i, C) \le \Gamma + 4\delta L.$$ The proof uses convexity of distance and $(I-T)^{-1}(1 - T\mathbf{1}) = \mathbf{1}$.

Label	Status	Statement and proof content
C2 lifetime to occupation measure	ready after review	For a finite substochastic T, either $$\|(I-T)^{-1}\mathbf{1}\|_\infty \le L$$ or there is a probability μ with $$\mu(1-T\mathbf{1}) \le 1/L, \qquad \|\mu T-\mu\|_1 \le 2/L.$$ The proof is the normalized occupation measure of a long-lived starting row.
C3 distance separator	ready with source caveat	If p_i is $H\tau$-far from a convex hull C, finite ℓ_1/ℓ_∞ separation gives affine ϕ with $\|\phi\|_* \le 1$ and $$\phi(p_i) \ge \sup_C \phi + H\tau.$$ Canonical use needs either an internal finite proof or a byte-verified source.
C4 high-slice drift	ready	If $\|p_i - \sum_j q_i(j)p_j\|_1 \le \varepsilon$, ϕ is L_ϕ-Lipschitz, $h_i = M - \phi(p_i)$, and $H_\gamma = \{h \le \gamma\}$, then $$q_i(H_\gamma^c) \le \frac{h_i + L_\phi\varepsilon}{\gamma}.$$ This is a one-line Markov inequality applied to $\sum_j q_i(j)h_j \le h_i + L_\phi\varepsilon$.
C5 high-slice bad closure	conditional	With bad-set exit $s_i = q_i(B^c)$, $$q_i((B\cap H_\gamma)^c) \le s_i + \frac{h_i + L_\phi\varepsilon}{\gamma}.$$ Rows in an $O(\delta)$ top core map into a $G\tau$ high slice with $O(\tau)$ leakage, provided their bad-set exit is $O(\tau)$.
C6 top-face exposure	ready	If u maximizes ϕ, and every row outside $B_1(u,\rho)$ has $\phi(u)-\phi(p_i) \ge \Delta_\phi$, then $$e_u(\rho) \ge \Delta_\phi/M_\phi, \qquad M_\phi = \max_i(\phi(u)-\phi(p_i)).$$ For $\|\phi\|_* \le 1$, $M_\phi \le 2+4\delta$.
C7 shadow edge	ready with global-max hypothesis	If u is a global ϕ-maximizer and $e_u(\rho) < \kappa$, then there is a row j with $\|p_j - u\|_1 \ge \rho$ and $$\phi(p_j) \ge \phi(u) - (2+4\delta)\kappa.$$ If u is $H\tau$-above C, then j remains $(H-(2+4\delta)k)\tau$-above C.

Label	Status	Statement and proof content
C8 shadow recurrence	conditional	For a substochastic shadow kernel S on finite H with row sums at least $1-\varepsilon_s$, the row-normalized chain has a recurrent class and a probability π with $$\|\pi S-\pi\|_1 \le \varepsilon_s.$$ If H is rowwise q-closed, then also $\pi q(H^c) \le \varepsilon_q$.
C10 failed-exposedness circuit	ready	If $e_v(\rho) < \kappa$, finite minimax and LP duality give $$\sum_{j\in S_v(\rho)} \mu_j(p_j - v) = \sum_i (\beta_i - \alpha_i)(p_i - v), \qquad \sum_i \beta_i < \kappa,$$ with complementarity support: α on the zero face and β on the top face. The α-mass is uncontrolled.

5.2 Small-Case And Model Results

Label	Status	Content
Rank ≤ 2, $n = 3$, simplex row polytopes	ready / exact	Segment and simplex barycentric coordinates show all vertices are κ-exposed at the required square-root scale for suitable constants, and every row lies exactly in their hull.
Explicit $n = 4$ quadrilateral family	ready / exact	For the family $P_t = I - uv^T$ with $\delta = t^2$, all four row vertices are well exposed at $\rho = t$. Gurobi verified the exposedness LP at $t = 1/10$ with objective 0.99.
General $n = 4$, rank 3, $2\|2$ circuit	ready at coefficient level	For a positive circuit $$ap_0 + bp_1 = cp_2 + dp_3, \qquad a+b=c+d=1,$$ the value assignments expose p_0, p_1, p_2, p_3 at levels b, a, d, c. If one coefficient is $< k\tau$, the controlled vertex is within $O(k\tau)$ of the opposite edge.
Robust-coordinate push-forward	ready conditional	If a stochastic kernel U reconstructs rows from representatives, then canonical coordinates pushed through U satisfy $$\operatorname{neg}(\lambda(p_i)) \le \operatorname{neg}(p_i) \le \delta.$$ The remaining need is $G = I + O(\delta)$ in row-ℓ_1 operator norm.

Label	Status	Content
Exact representative reconstruction implies $G = I$	ready	If $p_j = \sum_a U_{ja} r^a$ exactly and the representatives are affinely independent, then applying $r^b = r^b P$ forces $$G_{ba} = \sum_j r_j^b U_{ja} = \delta_{ba}.$$

5.3 C9 Campaign Results

Label	Status	Content
C9-A high-core pruning	ready with hypothesis	For $H_0 = B \cap \{h \leq A_0 \delta\}$, $H_1 = B \cap \{h \leq G\tau\}$, and a probability m on B, $$\nu q(H_1^c) \leq \eta / m(H_0) + ((A_0 + 4)/G)\tau,$$ $$\nu = m\|_{H_0} / m(H_0).$$ Thus sufficient high-core mass gives distributional q-closure into H_1 and rowwise pruning on most of ν.
C9-A fallback refutation	false	The statement "small $m(H_0)$ implies a Lyapunov/resolvent fallback" is false for an arbitrary quasi-stationary bad measure. A stochastic idempotent with two closed states, one at height 0 and one at height $G\tau/2$, can have $m(H_0) = 0$, zero exit, and no drift. The missing datum is return mass to the high core.
C9-B Step 3 leakage	ready and limiting	Step 3 gives only $$\mu(H_1^c) \leq (2 + 4\delta)k/G,$$ not $O(\tau)$, for arbitrary failed-exposedness witnesses. Therefore high-supported shadow witnesses require extra q-compatibility or an LP minimum-leakage condition.
C9-B q-leakage	ready	For the repaired kernel, $$q_i(H_1^c) \leq (c_0 + 4)\tau / G$$ when $h_i \leq c_0 \delta$. This is the correct q-side leakage estimate, but arbitrary shadow witnesses need not be q-side witnesses.

Label	Status	Content
C9-C Lyapunov package	ready	For a killed finite kernel, a bounded potential with drift $c\tau$ gives $$\|(I-T)^{-1}\|_{\infty\to\infty} = O(1/\tau).$$ Failure of this resolvent bound yields a quasi-stationary occupation measure with $O(\tau)$ exit and $O(\tau)$ stationarity defect.
C9-D distributional-to-rowwise refutation	false	The deterministic path $$q_i = \delta_{i+1}\ (i<N), \qquad q_N(B^c)=1, \qquad m_i = 1/N$$ has $mq(B^c) = 1/N$ and $\|mT-m\|_1 = 2/N$, but every nonempty subset has a row with full leakage. Thus distributional q-quasi-closure alone cannot produce a rowwise closed core.
C9-D pi-coupled replacement	conditional / open interface	The useful replacement is a common averaging law π satisfying $$\pi q(H^c) = O(\tau), \quad \|\pi S - \pi\|_1 = O(\tau), \quad \pi(1-S\mathbf{1}) = O(\tau).$$ The preferred route is $\pi = m$, the bad-kernel occupation law. Proving shadow witnesses approximately stationary for the same m is open.
C9-E repaired scoring	evidence	In 4473 scored direct $P = AB$, $BA = I$ reports, no C9 failure was found. Top threats had nonempty bad sets but $bad\_lifetime = 1$, zero shadow-leakage proxy, and q-exit in the benign one-step regime.
C9-F frozen LP diagnostic	evidence	The frozen-row model found no small-δ C9 failure. Hume and direct small-defect examples stop by exposed augmentation or no bad class. A regular 12-gon realizes the frozen failure pattern only at large negative mass $\delta \approx 0.205$.

6 False Shortcuts And Necessary Caveats

The session improved the proof state partly by finding statements that must not be used.

Warning 1 (Pure convex geometry is insufficient)**.** Dense polygon and cyclic-chain examples show that sequential pointwise deletion or local exposed-or-redundant reasoning can cycle. The proof must use $P^2 = P$ and $\operatorname{neg}(p_i) \le \delta$.

Warning 2 (One-scale high-slice closure is too strong)**.** The drift bound gives

$$q_i(H^c_{G\tau}) \le (h_i + 4\delta)/(G\tau).$$

Rows with $h_i = O(\tau)$ therefore give only constant leakage. The correct rowwise statement is two-scale:

$$O(\delta) \text{ top core} \longrightarrow G\tau \text{ high slice with } O(\tau) \text{ leakage.}$$

Warning 3 (Small high-core mass does not force drift)**.** As recorded in C9-A, an arbitrary quasi-stationary measure may live forever on a lower closed class. A fallback needs return information or a genuine Lyapunov potential, not only $m(H_0)$ being small.

Warning 4 (Distributional q-closure is not rowwise closure)**.** The deterministic path example of C9-D shows that

$$mq(B^c),\ \|mT - m\|_1 = O(\tau)$$

does not imply any nonempty subset with rowwise $O(\tau)$ exit.

Warning 5 (Shadow leakage is not q-drift)**.** Step 3/Step 5 witnesses are LP witnesses for failed exposedness. Unless they are coupled to q_i, their leakage below a high slice does not imply q-exit or a q-Lyapunov drift.

Warning 6 (Raw affine-circuit lower bound is false)**.** The stochastic idempotent with rows

$$(1, 0, 0), \quad (0, 1, 0), \quad (1/2, 1/2, 0)$$

has a separated affine circuit and zero negative mass. Step 7 must restrict to reduced row-vertex or Step-5 anchored circuits with explicit witness mass and residual.

7 Open Conjectural Interfaces

Open Problem 1 (C9: q/shadow interface)**.** Prove either a pi-coupled interface

$$\pi q(H^c) \le C\tau, \qquad \|\pi S - \pi\|_1 \le C\tau, \qquad \pi(1 - S\mathbf{1}) \le C\tau$$

with $\pi = m$ from the bad-kernel occupation law, or prove that failure of such q-compatible witness selection produces the C12 alpha-budget or calibrated-dual obstruction.

Open Problem 2 (C12: alpha budget or calibrated dual)**.** In a q-quasi-closed high bad class, choose failed-exposedness witnesses either with averaged alpha mass $M = O(1)$, or with a q-compatible calibration whose residual is $O(\tau)$ and whose separated witness mass θ is quantified. The known Step 5 LP duality only controls the β-mass; it does not control the lower-face α-mass.

Open Problem 3 (C13: separated-circuit negative-mass lower bound)**.** For the reduced or Step-5 anchored aggregate circuit output by C12, prove

$$\max_i \operatorname{neg}(p_i) \ge c\,\theta\rho - CE,$$

or the stronger mass-free version $c\rho^2 - CE$, with explicit dependence on separated witness mass θ and residual E. If Step 6 supplies only $\theta = \Omega(\tau)$, the weak $\theta\rho^2$ bound is not enough.

Open Problem 4 (Closed-bad-class augmentation)**.** Let B be a high bad row-vertex class farther than $D\tau$ from $C = \operatorname{conv} R_0$, enlarged by ρ-clusters, and suppose B is q-closed in the sense supplied by the long-lifetime alternative. Prove that B contains a row vertex w with

$$\operatorname{dist}_1(w, C) > 4\rho, \qquad e_w(\rho) \ge \kappa.$$

This is blocked exactly by C9, C12, and C13.

Open Problem 5 (Robust-coordinate interpolation upgrade)**.** Given a maximal square-root-exposed skeleton R and rowwise $\operatorname{conv} R$-reconstruction at $O(\tau)$, find a stochastic kernel U such that

$$\left\| p_i - \sum_a U_{ia} r^a \right\|_1 \leq C\tau$$

and the representative interpolation matrix satisfies

$$\max_b \sum_a \left| \sum_j r_j^b U_{ja} - \delta_{ba} \right| \leq C\delta.$$

8 Numerical And Computational Work

All numerics are evidence only. They were used to identify false alarms, constants, and candidate certificate shapes.

Artifact	Finding
`exposed_hull_metrics.py` and family analyzers	Compute row negative mass, row vertices, exposedness LPs, $W_{\rho,\kappa}$, and distance to $\operatorname{conv} W$.
Hume and Hume products	Sharp for the $\sqrt{\delta}$ exponent but not exposed-hull obstructions. At tested square-root scales the relevant vertices remain well exposed and rows lie in their hull.
Similarity-conjugated stochastic idempotents	No serious small-defect threat found. Missing well-exposed vertices were very close to the hull of the remaining well-exposed vertices.
Direct exact $P = AB$, $BA = I$, $P\mathbf{1} = \mathbf{1}$ search	No counterexample found. Aggressive constants $\rho = 0.5\tau$, $\kappa = 0.5\tau$ can drop anchor vertices, but the same samples become harmless for $\rho \geq \tau$. Conservative worst finite ratio was about 0.1985.
Small cases and Gurobi LP	Rank ≤ 2, $n = 3$, simplex row polytopes, and an explicit $n = 4$ quadrilateral are benign. Gurobi 13.0.2 solved an exposedness LP with the expected objective 0.99.
Stress families	Dense polygons and cyclic chains explain why sequential deletion fails, but exact polygon projections have constant negative mass and local small-negative stencils have order-one idempotency defect.
LP/game frozen diagnostic	The full negation is bilinear/nonconvex before freezing P. Frozen LPs separate Hume augmentation from the large-negative-mass regular-polygon warning.
C9 repaired scorer	In direct samples, top C9 threats exit in one step or have high-supported shadow witnesses. No small-defect long-lived bad trap was seen.

9 Literature Search

The literature search found no theorem equivalent to Conjecture 1. The useful sources are indirect and must be acquired into local `refs/` before canonical use.

Highest-priority leads are:

- Blackwell on idempotent Markoff chains.
- Schwarz on semigroups of stochastic matrices.
- Gonzalez–Hartfiel and Gonzalez–Torres on exact stochastic idempotent matrix spaces and cores.
- Hoffman and modern Hoffman-constant literature for LP error bounds.
- Choquet–Corson–Klee and Straszewicz for exposed-point background.
- Agaev–Chebotarev, Seneta, and related Markov projector/perturbation work for comparison language around resolvents.

The assessment is negative as a proof source: exact stochastic idempotent structure helps identify the endpoint after the skeleton theorem, but the dimension-free signed-retraction $\sqrt{\delta}$ stability estimate appears to be new.

10 Promisingness Assessment

10.1 Why The Primary Route Is Promising

The primary route is promising for four concrete reasons.

1. The positive-coordinate repair and bad-kernel resolvent half are small, deterministic, and proof-ready.
2. All stress tests and exact searches so far fail to produce a small-δ counterexample. The false alarms are explained by bad constants, constant negative mass, or loss of exact idempotency.
3. The closed-bad-class problem has been decomposed into a finite list of small contracts. The blocker cut is now exactly $\{C9, C12, C13\}$, not an amorphous gap.
4. The latest C9 work did not merely fail; it identified the correct replacement interface: use a common law $\pi = m$ for q-closure and shadow aggregation, or turn the incompatibility into C12 data.

10.2 Why The Route Is Still Risky

The route is still genuinely open. The risks are:

1. C12 may be hard: failed-exposedness LP duality leaves uncontrolled zero-face α-mass, and q-quasi-closedness does not automatically see that mass.
2. C13 needs a reduced or anchored circuit lower bound. The raw affine circuit statement is false, so the exact normalization matters.
3. C9 now depends on a coupled witness-selection theorem. Step 3 alone cannot supply the desired leakage.

4. Generic Hoffman/convex-geometry black boxes have conditioning problems; the proof must exploit $P^2 = P$ and row negative mass directly.

10.3 Recommended Next Work

The highest-leverage next tasks are:

1. Attack C12 directly by minimizing averaged α-mass in the failed-exposedness LPs and extracting the dual obstruction.
2. Build a fixed-combinatorics LP/nonlinear model for the C13 medium bound

$$\delta \geq c\,\theta\rho - CE$$

and search for small rational counterexamples before formalization.

3. Reformulate C9 as a $\pi = m$ witness-selection problem:

$$\|mS - m\|_1 = O(\tau), \qquad m(1 - S\mathbf{1}) = O(\tau),$$

using q-compatible or minimum-leakage witnesses.

4. Package C0, C1, C2, C4, C5, C6, C7, C8, and C10 for Agent A review as isolated candidate contracts.
5. Continue the robust-coordinate route only as a certificate-mining parallel track; its exact interpolation lemma may reduce to the same closed-bad-class dichotomy.

11 Artifact Ledger

The main session artifacts used for this report are:

- `agent-B/notes/op-exposed-hull-mission-control.md`
- `agent-B/notes/op-exposed-hull-session-2026-06-07.md`
- `docs/plans/2026-06-07-op-exposed-hull-attack-plan.md`
- `docs/plans/2026-06-07-op-exposed-hull-hard-block-delegation.md`
- `docs/plans/2026-06-07-op-exposed-hull-blocker-subplans.md`
- `agent-B/notes/subagent-op-exposed-hull-lp-dual.md`
- `agent-B/notes/subagent-op-exposed-hull-skeleton.md`
- `agent-B/notes/subagent-op-exposed-hull-bad-kernel.md`
- `agent-B/notes/subagent-op-exposed-hull-robust-coordinates.md`
- `agent-B/notes/subagent-op-exposed-hull-interpolation-upgrade.md`
- `agent-B/notes/subagent-op-exposed-hull-computational.md`
- `agent-B/notes/subagent-op-exposed-hull-small-cases.md`

- agent-B/notes/subagent-op-exposed-hull-stress-tests.md
- agent-B/notes/subagent-op-exposed-hull-direct-search.md
- agent-B/notes/subagent-op-exposed-hull-lp-game.md
- agent-B/notes/subagent-op-exposed-hull-n4-circuit.md
- agent-B/notes/subagent-op-exposed-hull-step1-high-slice.md
- agent-B/notes/subagent-op-exposed-hull-step2-top-face-exposure.md
- agent-B/notes/subagent-op-exposed-hull-step3-shadow-edge.md
- agent-B/notes/subagent-op-exposed-hull-step4-shadow-recurrence.md
- agent-B/notes/subagent-op-exposed-hull-step5-circuit-extraction.md
- agent-B/notes/subagent-op-exposed-hull-step6-circuit-aggregation.md
- agent-B/notes/subagent-op-exposed-hull-step7-circuit-lower-bound.md
- agent-B/notes/subagent-op-exposed-hull-step8-capstone-packaging.md
- agent-B/notes/subagent-op-exposed-hull-c9a-high-core-pruning.md
- agent-B/notes/subagent-op-exposed-hull-c9b-shadow-leakage.md
- agent-B/notes/subagent-op-exposed-hull-c9c-lyapunov-fallback.md
- agent-B/notes/subagent-op-exposed-hull-c9d-markov-coupling.md
- agent-B/notes/subagent-op-exposed-hull-c9e-computational-scoring.md
- agent-B/notes/subagent-op-exposed-hull-c9f-counterexample-lp.md
- agent-B/notes/subagent-op-exposed-hull-frameworks.md
- agent-B/notes/subagent-op-exposed-hull-literature.md
- agent-B/notes/op-exposed-hull-literature-scout-2026-06-07.md
- agent-B/experiments/op-exposed-hull/

12 Bottom Line

The classical path is promising, but the rigorous proof has not landed. The session converted a broad search problem into a precise proof program. The primary route should now be judged by C9/C12/C13:

$$\begin{aligned}&\text{pi-coupled q/shadow closure}\\&\quad+\ \text{alpha-budget/calibrated aggregation}\\&\quad+\ \text{anchored separated-circuit lower bound.}\end{aligned}$$

If these three statements are true with the stated scales, the remaining capstone is short. If any one fails, the existing LP/game diagnostics are well positioned to produce a concrete counterexample candidate or a revised interface. At this checkpoint, no numerical or small-case evidence points to a small-defect counterexample; the main difficulty is analytic, not experimental.